

Name: Niel Jhon E. Celocia

Course: BSIT-4

Capture the Flag

Last mission, you built reusable Angular components for the Student Portal.

Unfortunately, you hardcoded every piece of information directly into the HTML. Every student's profile looks identical, and announcements can only be changed by editing the HTML.

Your mission is to **bring the portal to life using Angular Interpolation** (`{{ }}`).

Note from the instructor before you start:

These activities involve using your own logic. We are strictly to implement NOT TO USE AI to generate the logic for you. You may only use AI for RESEARCHING the syntax or UI, but NEVER THE LOGIC.

Consequence: Immediate 0 score in your grade for this activity.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/interpolation`

Capture

- Show your current Git branch.

Flag 1 – Escape the Hardcoded Trap

Open your **Student Profile Card**.

Right now, your HTML probably looks something like this:

```
<h2>Juan Dela Cruz</h2>
<p>BSIT</p>
<p>3rd Year</p>
<p>2026-0001</p>
```

Your mission is to remove the hardcoded values.

Move them into your **TypeScript** component instead. (".ts")

Hint

```
{{ variableName }}
```

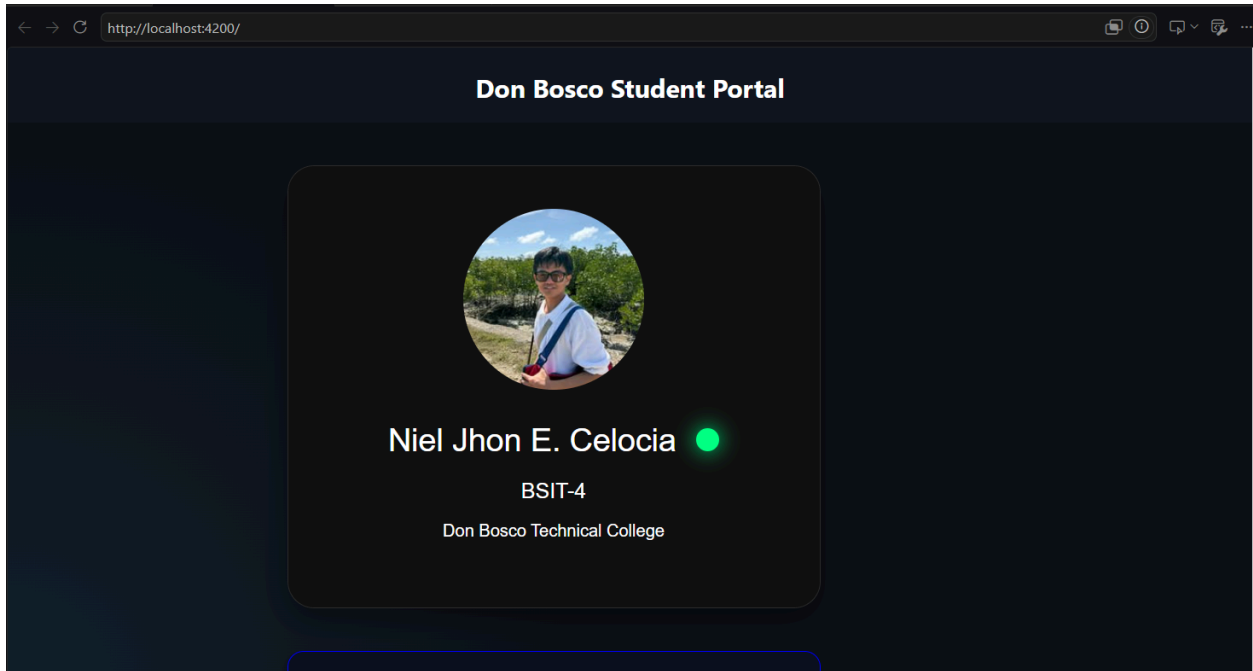
Your HTML should no longer contain actual student information. It should only reference variables.

Your profile card should display:

- Student Name
- Student Number
- Course
- Year Level
- Section
- Nickname (all lowercase)
- Email
- Grade

Capture

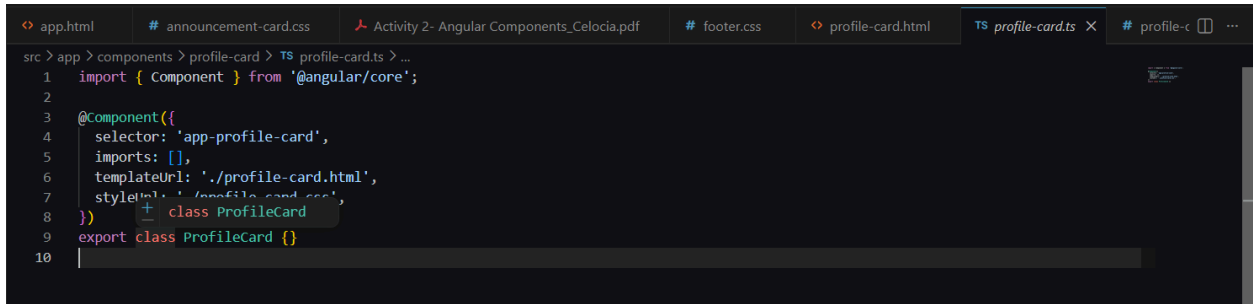
- HTML file
- TypeScript file
- Browser



```
app.html # announcement-card.css Activity 2- Angular Components_Celocia.pdf # footer.css profile-card.html X # profile-card.css # af 11
src > app > components > profile-card > profile-card.html > ...
Go to component
1 <div class="profile-card">
2
3   
8
9
10  <h1>
11    Niel Jhon E. Celocia
12    <span class="online"></span>
13  </h1>
14
15  <h3>BSIT-4</h3>
16
17  <p>Don Bosco Technical College</p>
18
19
20 </div>
21
22

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + ~ 11 12 13 14 15 16 17 18 19 20 21 22
PS C:\Users\nielj\my-first-angular-app> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\nielj\my-first-angular-app>
```



```
src > app > components > profile-card > TS profile-card.ts > ...
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-profile-card',
5    imports: [],
6    templateUrl: './profile-card.html',
7    styleUrls: ['./profile-card.css'],
8  })
9  export class ProfileCard {}
10
```

Flag 2 – Dynamic Profile Card

Verify that **every** piece of information in your profile card now comes from TypeScript.

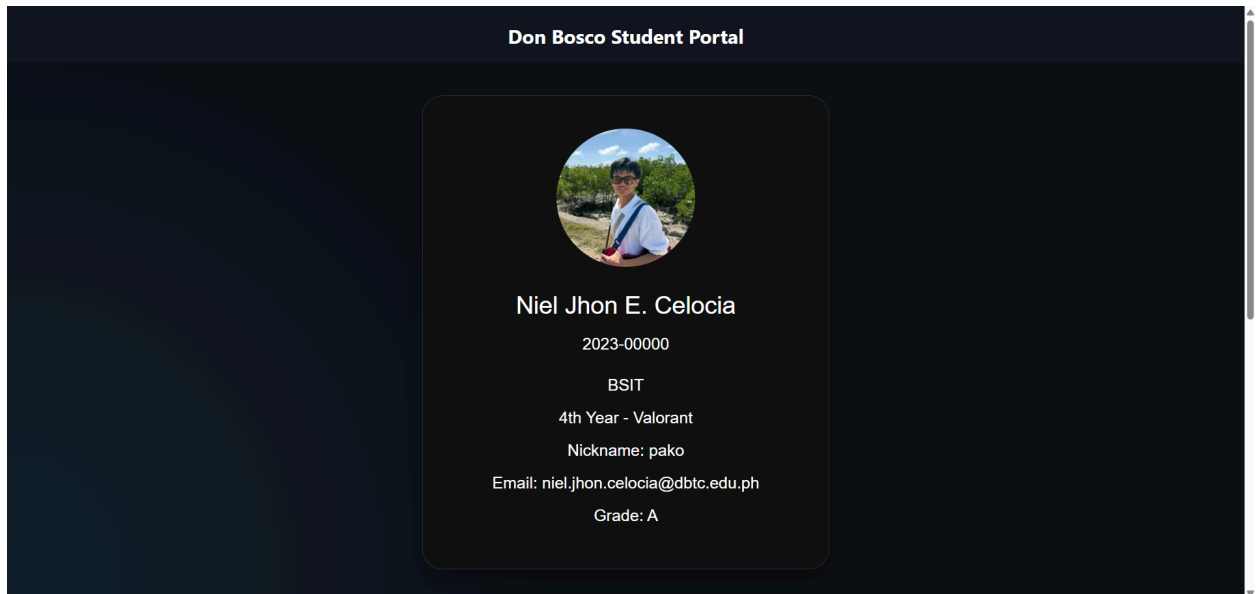
If changing a value in the TypeScript file updates the browser without touching the HTML, you've captured the flag.

Capture

- Updated TypeScript file with changed value

```
TS profile-card.ts M X TS profile-card.spec.ts <> profile-card.html M MyF
src > app > components > profile-card > TS profile-card.ts > ...
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-profile-card',
5    imports: [],
6    templateUrl: './profile-card.html',
7    styleUrls: ['./profile-card.css'],
8  })
9  export class ProfileCard {
10    StudentName: string = 'Niel Jhon E. Celocia';
11    StudentNumber: string = '2023-00000';
12    Course: string = 'BSIT';
13    YearLevel: string = '4th Year';
14    Section: string = 'Valorant';
15    Nickname: string = 'pako';
16    Email: string = 'niel.jhon.celocia@dbtc.edu.ph';
17    Grade: string = 'A';
18
19  }
20
21
```

- Browser



Flag 3 – Dynamic Announcements

Your announcement component is also hardcoded.

Instead of writing directly in HTML, create variables for each announcement.

Example idea:

announcementTitle1
announcementBody1

announcementTitle2
announcementBody2

announcementTitle3
announcementBody3

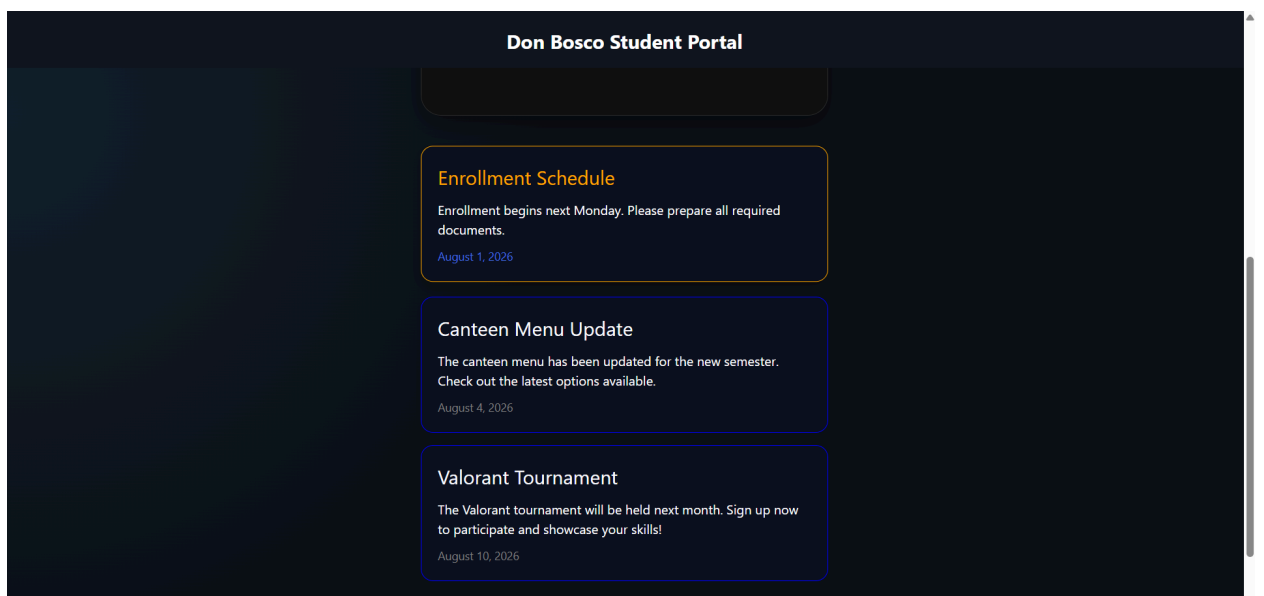
Use interpolation so every announcement card gets its content from TypeScript.

💡 Imagine these values came from a JSON file or API in the future.

📸 Capture


```
src > app > components > announcement-card > TS announcement-card.ts > ...
3  @Component({
4      selector: 'app-announcement-card',
5      imports: [],
6      templateUrl: './announcement-card.html',
7      styleUrls: ['./announcement-card.css'],
8  })
9  export class AnnouncementCard {
10     announcementTitle1: string = 'Enrollment Schedule';
11     announcementBody1: string = 'Enrollment begins next Monday. Please prepare all required documents.';
12     announcementDate1: string = 'August 1, 2026';
13
14     announcementTitle2: string = 'Canteen Menu Update';
15     announcementBody2: string = 'The canteen menu has been updated for the new semester. Check out the latest options available.';
16     announcementDate2: string = 'August 4, 2026';
17
18     announcementTitle3: string = 'Valorant Tournament';
19     announcementBody3: string = 'The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!';
20     announcementDate3: string = 'August 10, 2026';
21  }
22
23
```

-
- Browser



Flag 4 – Thinking Like Data

Now add 7 more cards to make the announcements a total of 10 cards.

Capture

- Show your ten announcement cards in Browser

Enrollment Schedule

Enrollment begins next Monday. Please prepare all required documents.

August 1, 2026

Canteen Menu Update

The canteen menu has been updated for the new semester. Check out the latest options available.

August 4, 2026

Valorant Tournament

The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!

August 10, 2026

Library Renovation

The library will be closed for renovation from August 15 to September 15. Please plan your visits accordingly.

Don Bosco Student Portal

September 15. Please plan your visits accordingly.

August 12, 2026

New Course Offering

We are excited to announce a new course offering in Artificial Intelligence starting this semester. Enroll now to be part of this innovative program.

August 20, 2026

Sports Day

Join us for the annual Sports Day event on September 5. Participate in various sports activities and enjoy a day of fun and fitness!

August 25, 2026

Guest Lecture Series

We are hosting a series of guest lectures from industry experts throughout the semester. Stay tuned for the schedule and topics.

August 30, 2026

Don Bosco Student Portal

August 30, 2026

Community Service Project

Get involved in our community service project this semester. Opportunities are available for volunteering and making a positive impact.

September 5, 2026

Tech Fair

The annual Tech Fair will be held on September 15. Explore the latest technological innovations and projects from our students.

September 10, 2026

End of Semester Celebration

Join us for the end of semester celebration on September 30. Celebrate the achievements of our students and faculty with fun activities and performances.

September 25, 2026

-
- TypeScript file

```

src > app > components > announcement-card > ts announcement-card.ts > AnnouncementCard
1  import { Component } from '@angular/core';
2
3
4  @Component({
5    selector: 'app-announcement-card',
6    imports: [],
7    templateUrl: './announcement-card.html',
8    styleUrls: ['./announcement-card.css'],
9  })
10 export class AnnouncementCard {
11   announcementTitle1: string = 'Enrollment Schedule';
12   announcementBody1: string = 'Enrollment begins next Monday. Please prepare all required documents.';
13   announcementDate1: string = 'August 1, 2026';
14
15   announcementTitle2: string = 'Canteen Menu Update';
16   announcementBody2: string = 'The canteen menu has been updated for the new semester. Check out the latest options available.';
17   announcementDate2: string = 'August 4, 2026';
18
19   announcementTitle3: string = 'Valorant Tournament';
20   announcementBody3: string = 'The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!';
21   announcementDate3: string = 'August 10, 2026';
22
23   announcementTitle4: string = 'Library Renovation';
24   announcementBody4: string = 'The library will be closed for renovation from August 15 to September 15. Please plan your visits accordingly.';
25   announcementDate4: string = 'August 12, 2026';
26
27   announcementTitle5: string = 'New Course Offering';
28   announcementBody5: string = 'We are excited to announce a new course offering in Artificial Intelligence starting this semester. Enroll now to be part of this innovative program.';
29   announcementDate5: string = 'August 20, 2026';
30
31   announcementTitle6: string = 'Sports Day';
32   announcementBody6: string = 'Join us for the annual Sports Day event on September 5. Participate in various sports activities and enjoy a day of fun and fitness!';
33   announcementDate6: string = 'August 25, 2026';
34
35   announcementTitle7: string = 'Guest Lecture Series';
36   announcementBody7: string = 'We are hosting a series of guest lectures from industry experts throughout the semester. Stay tuned for the schedule and topics.';
37   announcementDate7: string = 'August 30, 2026';
38
39   announcementTitle8: string = 'Community Service Project';
40   announcementBody8: string = 'Get involved in our community service project this semester. Opportunities are available for volunteering and making a positive impact.';
41   announcementDate8: string = 'September 5, 2026';
42
43   announcementTitle9: string = 'Tech Fair';
44   announcementBody9: string = 'The annual Tech Fair will be held on September 15. Explore the latest technological innovations and projects from our students.';
45   announcementDate9: string = 'September 10, 2026';
46
47   announcementTitle10: string = 'End of Semester Celebration';
48   announcementBody10: string = 'Join us for the end of semester celebration on September 30. Celebrate the achievements of our students and faculty with fun activities and performances.';
49   announcementDate10: string = 'September 25, 2026';
50 }
51
52

```

Notice how your announcement variables are starting to repeat.

This is normal, but not ideal.

Question to think about (you don't need to answer, just think):

If there were 100 announcements, would creating 200 variables be a good idea?

(You'll solve this in a future lesson.)

Flag 5 – Academic Standing

Your profile already displays the student's grade.

Now make the portal smarter.

Below the grade, display:

- **Pass**

- **Fail**

depending on the grade.

Hint

Angular interpolation can evaluate expressions.

{{ condition ? "True Value" : "False Value" }}

Grading condition: A student can only pass if their grade is 1.0 to 3.0.

Capture

- Interpolation condition

```
Grade: number = 2.5;

get status(): string {
  return (this.Grade >= 1.0 && this.Grade <= 3.0) ? 'Pass' : 'Fail';
}
```

-
- Browser



Niel Jhon E. Celocia

2023-00000

BSIT

4th Year - Valorant

Nickname: pako

Email: niel.jhon.celocia@dbtc.edu.ph

Grade: 2.5

Status: Pass

Flag 6 – Formatting Text

Your student's nickname is stored in lowercase.

Example:

juan

Display it in uppercase without changing the value of the variable.

Hint

Interpolation can call simple methods.

Expected:

JUAN

Capture

- Method used to display it in uppercase

```
<h1>
|   {{ StudentName.toUpperCase() }}
</h1>
```

-
- Browser



NIEL JHON E. CELOCIA

2023-00000

BSIT

4th Year - Valorant

Nickname: pako

Email: niel.jhon.celocia@dbtc.edu.ph

Grade: 2.5

Status: Pass

Flag 7 – School Email

Every student has the same format of email in Don Bosco.

Example:

juan.delacruz@dbtc-cebu.edu.ph

Instead of storing the full email, remove the “email” variable, then construct the email inside the template, by using the student’s name as variable, concatenated with the email suffix.

No extra variable should be created.

Capture

- Your logic to concatenate the name and email suffix using interpolation

```
</h2>
<h3>{{ Course }}</h3>
<h3>{{ YearLevel }} - {{ Section }}</h3>
<h3>Nickname: {{ Nickname }}</h3>
<p>Email: {{ StudentFirstName.toLowerCase() + '.' + StudentLastName.toLowerCase() + '@dbtc-cebu.edu.ph' }}</p>
<h3>Grade: {{ Grade }} </h3>
<p>Status: {{ status }}</p>
```

-
- Browser



NIEL JHON E. CELOCIA

2023-00000

BSIT

4th Year - Valorant

Nickname: pako

Email: niel.jhon.celocia@dbtc-cebu.edu.ph

Grade: 2.5

Status: Pass

Flag 8 – Copyright Footer

Every website needs a footer.

Create a variable:

`currentYear`

Display:

© 2026 Student Portal

using interpolation.

Example idea:

© {{ currentYear }}

Capture

- Method used to display the currentYear without hardcoding “2026”

```
src > app > components > footer > TS footer.ts > ...
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-footer',
5    imports: [],
6    templateUrl: './footer.html',
7    styleUrls: ['./footer.css'],
8  })
9  export class Footer {
10   currentYear: number = new Date().getFullYear();
11 }
12
```

-
- Browser

Join us for the end of semester celebration on September 30.
Celebrate the achievements of our students and faculty with fun
activities and performances.
September 25, 2026

© 2026 Student Portal.

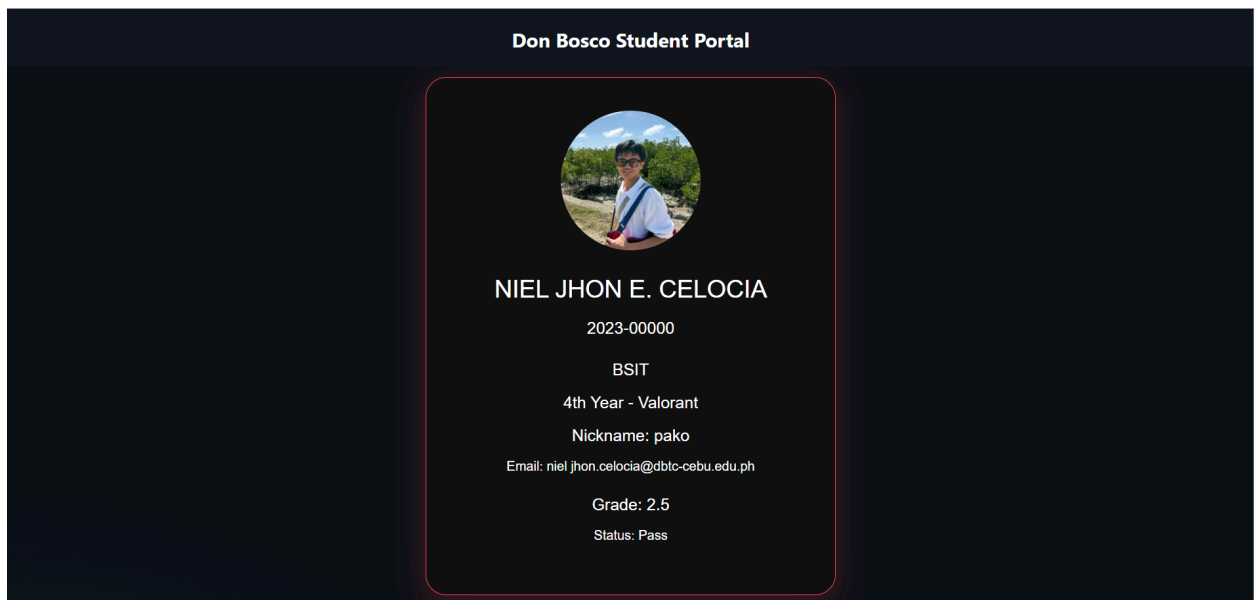
-

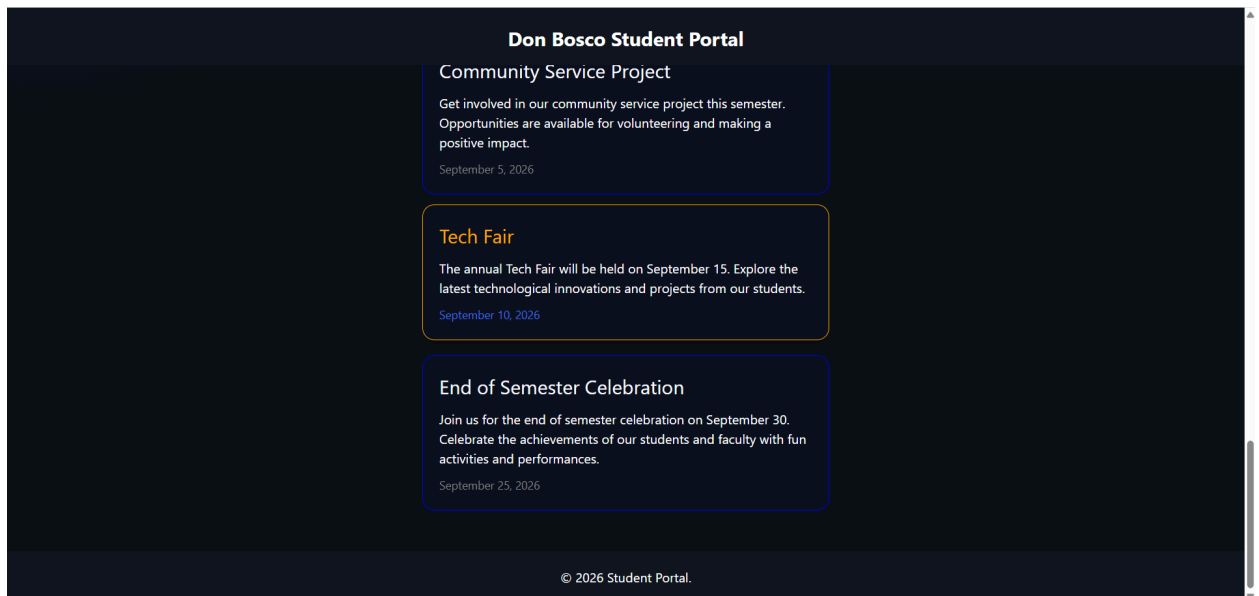
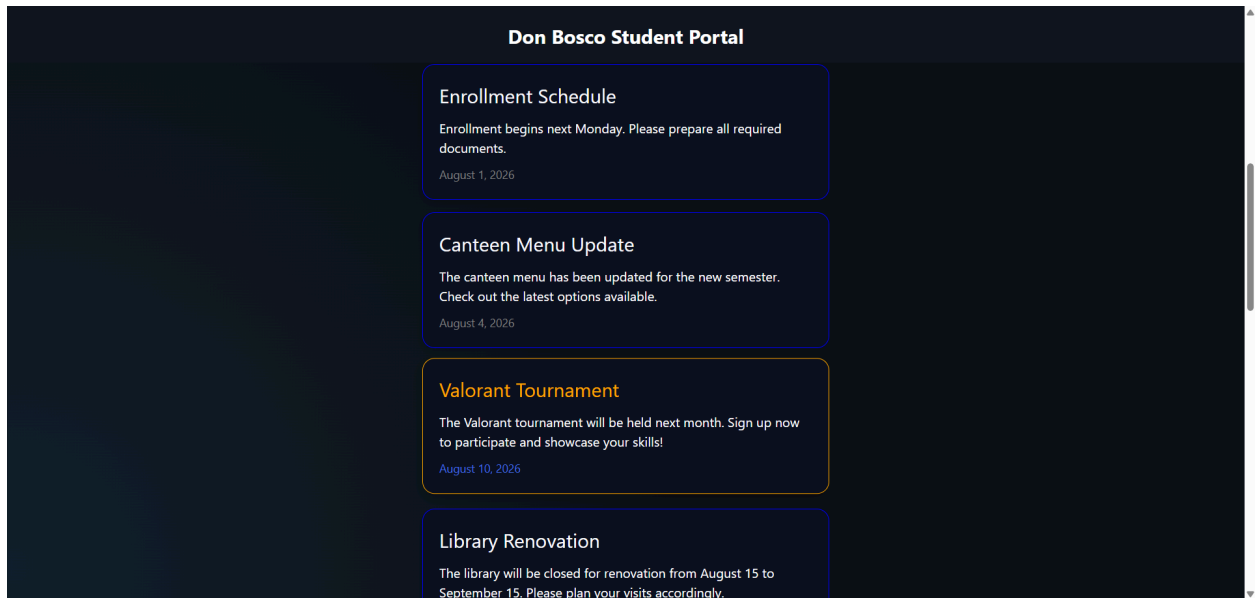
Imagine the year changes from 2026 to 2027. If you hardcoded the year, you will have to change this value yearly. You need to have a function for that.

Flag 9 – Commit Your Progress

📷 Final Flag Screenshot

- The entire completed browser UI for this activity





-
- Successful push to **feature/interpolation**

```
PS C:\Users\nielj\my-first-angular-app> git push origin feature/interpolation
Enumerating objects: 43, done.
Counting objects: 100% (43/43), done.
Delta compression using up to 12 threads
Compressing objects: 100% (29/29), done.
Writing objects: 100% (29/29), 4.68 KiB | 239.00 KiB/s, done.
Total 29 (delta 15), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (15/15), completed with 8 local objects.
remote:
remote: Create a pull request for 'feature/interpolation' on GitHub by visiting:
remote:   https://github.com/ejannn/ITFE403A\_Celocia/pull/new/feature/interpolation
remote:
To https://github.com/ejannn/ITFE403A_Celocia.git
 * [new branch]      feature/interpolation -> feature/interpolation
PS C:\Users\nielj\my-first-angular-app>
```